



Size of the 2026 Ebola outbreak and risk of cross-border spillover from Bundibugyo virus in Ituri Province, DR Congo, and its implications for preparedness: a recalibrated stochastic modelling study

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Summary

Background On May 15, 2026, WHO declared a Bundibugyo virus (BDBV) outbreak in Ituri Province, DR Congo with an estimated index case on April 1, 2026 (6-week pre-declaration interval). By May 24, DR Congo reported 906 suspected cases (105 confirmed and ten confirmed deaths) across three provinces. Uganda reported seven confirmed cases (three imported; four locally acquired including three health-care workers; case-fatality ratio 14%). WHO declared a Public Health Emergency of International Concern on May 17, 2026; Africa Centres for Disease Control and Prevention declared a Public Health Emergency of Continental Security on May 18, 2026. The study aimed to establish a short-term trajectory of the BDBV outbreak and probability of cross-border spillover into countries with elevated risk of importation to guide preparedness priorities.

Methods We calibrated a stochastic SEIRD (susceptible, exposed, infectious, recovered, and dead) ensemble model to the laboratory-confirmed case series, anchoring on 598 cumulative confirmed cases on June 8, 2026 (day 68) using simulation filtering (calibration window $\pm 30\%$; reporting fraction 1.0 for laboratory-confirmed cases). The case-fatality ratio was drawn from a previous value centred on the observed confirmed-case ratio of approximately 19% (115 of 598). A linked daily-hazard spillover model estimated importation probability for Uganda, South Sudan, Rwanda, and Burundi over a 12-week horizon. The early suspected-case series, which peaked at 1077 on May 26, 2026, before being substantially revised downward by laboratory reclassification, is reported for context but was not used for calibration.

Findings Laboratory-confirmed DR Congo cases rose from 33 on May 18, 2026, to 598 by June 8, 2026. Calibrated to the confirmed-case anchor (598 on June 8, 2026; central basic reproduction number $[R_0]=1.71$), the confirmed-case trajectory is most consistent with the central scenario. Under the central scenario the ensemble projected a median of 990 cumulative confirmed cases by week 12 (June 24, 2026; 90% prediction interval [PI] 709–1293) and 174 deaths; the low scenario projected 870 confirmed cases (90% PI 641–1133) and 160 deaths. The early suspected-case count (peak 1077 on 26 May 2026) was substantially revised by laboratory reclassification and is reported for context only. Cross-border spillover remained material: Uganda 94.2% importation probability (19 confirmed cases as of June 4, 2026, including five health-care worker infections and two deaths); South Sudan 69.3%; Rwanda 8.6%; and Burundi 2.0%. As of June 22, 2026, DR Congo has 1048 confirmed cases and 267 confirmed deaths and Uganda has 20 confirmed cases, two confirmed deaths, and one probable death. These numbers are changing daily and are likely to align with what is predicted in the central scenario.

Interpretation From the most recent laboratory-confirmed data, the outbreak is closer to what is predicted by the central scenario, even with the intensified response within DR Congo. However, uncertainty remains around reported case numbers due to low rate of contact tracing. Sustained control nonetheless remains the primary determinant of regional risk: importation into Uganda is already established, and South Sudan must continue to reinforce infection prevention and control, rapid response capacity, and cross-border surveillance under International Health Regulations 2005. These projections should be interpreted as exploratory preparedness-oriented estimates derived from a stochastic scenario-based modelling framework, rather than predictions generated from formally fitted epidemiological models using comprehensive parameter estimation and identifiability analyses.

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Introduction

Ebola virus disease remains one of the most consequential epidemic threats in sub-Saharan Africa because of its

high mortality, potential for rapid spread, and substantial health-system disruption. DR Congo has experienced multiple Ebola outbreaks over the past five decades, with

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For the French translation of the abstract see Online for appendix 1

For the Swahili translation of the abstract see Online for appendix 2

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Research in context

Evidence before this study

We searched PubMed and the WHO Disease Outbreak News archive on May 19, 2026, for outbreak-modelling studies of Bundibugyo virus (BDBV) published from database inception to May 19, 2026, using the terms "Bundibugyo", "ebolavirus", "spillover", "cross-border", AND "stochastic model". Previous outbreaks in Uganda (2007) and DR Congo (2012) suggested lower transmissibility and case-fatality ratio than Zaire ebolavirus, with reproduction number estimates ranging from 1.2 to 2.6. Real-time cross-border spillover modelling for BDBV has been scant, and no published study has integrated regional preparedness scoring with epidemic projections.

Added value of this study

This study combines epidemic modelling with spillover estimation to quantify regional risks associated with the 2026 Ituri outbreak using laboratory-confirmed case data from WHO Situation Reports (33 cases as of May 18, 2026, to 598 cases as of June 8, 2026), with projections extending 12 weeks from May 18 to Aug 10, 2026. It also integrates operational preparedness considerations relevant for neighbouring countries.

Implications of all the available evidence

Preparedness planning must urgently prioritise strengthening border surveillance, infection prevention and control, rapid response teams, and cross-border coordination mechanisms in countries at highest risk. Enhanced surveillance at formal and informal points of entry is essential to detect suspected cases early and prevent silent transmission. Infection prevention and control measures should be reinforced in front-line health facilities to reduce nosocomial spread and protect health-care workers. Rapid response teams must be trained, equipped, and pre-positioned for immediate deployment. Finally, robust cross-border coordination for South Sudan requires urgent reinforcement of infection prevention and control, rapid response, and point of entry surveillance. Control measures within DR Congo remain the primary determinant of regional spillover risk. Cross-border coordination under International Health Regulations 2005 and the Africa Centres for Disease Control and Prevention Public Health Emergency of Continental Security framework is crucial to harmonise detection and response.

repeated outbreaks highlighting the persistent interaction between zoonotic spillover, population displacement, and fragile public health infrastructure.^{1,2}

On May 15, 2026, WHO Regional Office for Africa received notification of a confirmed Bundibugyo virus (BDBV; *Orthoebolavirus bundibugyoense*) outbreak in Ituri Province, with sequence confirmation by the Institut National de Recherche Biomédicale. Retrospective investigation indicated the estimated index case occurred approximately on April 1, 2026, implying a 6-week pre-declaration interval during which transmission chains accumulated undetected. The WHO Director-General declared a Public Health Emergency of International Concern (PHEIC) on May 17, 2026; Africa Centres for Disease Control and Prevention (Africa CDC) declared a Public Health Emergency of Continental Security (PHECS) on May 18, 2026.^{3,4} WHO External Situation Report 01 (as of May 18, 2026) documented 528 cumulative suspected cases, 132 deaths, and 35 laboratory-confirmed cases (35 [6.6%] of 528) across DR Congo and Uganda.⁵

BDBV was first identified during the 2007 outbreak in western Uganda.⁶ Compared with Ebola virus, BDBV has historically demonstrated lower case-fatality ratios and lower transmissibility.^{7,8} Nevertheless, clinical manifestations and operational response requirements remain similar.⁹ The absence of a licensed vaccine specifically targeting BDBV also complicates preparedness and response strategies.

The current outbreak emerged in Ituri Province, characterised by insecurity, population displacement, and extensive cross-border mobility with Uganda and

South Sudan via the Mahagi–Goli–Paidha and Aru corridors. A 6-week interval between the estimated index case (April 1, 2026) and laboratory confirmation (May 15, 2026) substantially enlarged the infectious pool available for spillover before any response was initiated.^{10,11} Cross-border movement between eastern DR Congo, Uganda, and South Sudan occurs through both formal and informal routes, with many individuals crossing borders daily for trade, health care, and family visits. Such mobility substantially increases the likelihood of regional spillovers. By May 24, DR Congo reported 906 suspected cases (105 confirmed and ten confirmed deaths) across three provinces. Uganda reported seven confirmed cases (three imported; four locally acquired including three health-care workers; case-fatality ratio 14%).

The objective of this study was to estimate the short-term epidemic trajectory of the 2026 BDBV outbreak and quantify the probability of cross-border spillover into neighbouring countries, to support preparedness prioritisation among countries at elevated risk of importation, and to inform regional surveillance and rapid response planning under conditions of substantial uncertainty.

Methods

Study design

We conducted a stochastic modelling study using a discrete-time SEIRD (susceptible, exposed, infectious, recovered, and dead) framework, recalibrated to cumulative reported cases in Ituri Province on May 18, 2026 (day 48) and projected forward over a

24-week horizon to Sept 16, 2026. The 24-week horizon spans historical BDBV outbreak durations (2007 Uganda: approximately 3 months; 2012 DR Congo: approximately 2 months) and accommodates an operationally meaningful preparedness planning window.

The Preparedness Data Exchange platform

The modelling was conducted within the WHO Africa Office (AFRO) Preparedness Data Exchange (PDX) platform, the regional epidemic intelligence system serving all 47 member states of the African region.¹² The stochastic SEIRD model is implemented within the PDX predictions dashboard; the spillover model within the One Health module of PDX. All projections passed through the platform's human-decided governance tier before reporting. Analysis used Python version 3.11 (PDX platform) and R version 4.3.

Transmission model and scenario assumptions

We modelled transmission within a homogeneously mixing catchment of 350 000 (seven affected health zones, including Ituri and North Kivu). Spatial heterogeneity is not captured; this is a stated limitation.

Daily transitions were implemented as binomial draws using standard mass-action incidence (full mathematical specification in appendix 3 p 2). No vaccinated compartment was included as there is no licensed BDBV-specific vaccine.

Three transmissibility scenarios (low basic reproduction number [R_0]=1.42; central R_0 =1.71; high R_0 =2.08) spanned published BDBV estimates. Full model parameters are given in appendix 3 (pp 3–4). Response-intensity reductions applied from May 15, 2026, were 20% (low), 25% (central), and 15% (high). These distributions were chosen to span the range of published BDBV estimates from the 2007 Uganda and 2012 DR Congo outbreaks.^{6,8} The latent period was fixed at 9.7 days and the infectious period at 5.0 days, in line with the broader filovirus literature.^{13,14} The case-fatality ratio was drawn from a uniform distribution (0.15–0.25; (appendix 3 pp 3–4), centred on the case-fatality ratio observed among laboratory-confirmed cases in the current outbreak (115 [approximately 19%] of 598 as of June 8, 2026) and consistent with the historical BDBV range outbreaks.^{7,8} Because the analysis is anchored on laboratory-confirmed cases, the reporting fraction was set at 1.0 (laboratory-confirmed cases are, by definition, ascertained); the earlier suspected-case reporting fraction assumption is no longer required.

Calibration

Calibration was performed by simulation-filtering (approximate rejection sampling; not formal Bayesian inference); replicates were accepted if cumulative laboratory-confirmed cases at day 68 (June 8, 2026) fell within $\pm 30\%$ of 598 (range 419–777). We retained

400 calibration-passing replicates per scenario. The $\pm 30\%$ tolerance was chosen pragmatically given residual data uncertainty. The full calibration workflow is given in appendix 3 (p 4).

Data sources and provisional nature of inputs

Case data were taken from the WHO AFRO External Situation Reports and the WHO AFRO internal tracker through to June 8, 2026. The laboratory-confirmed case series was used as the primary calibration metric: confirmed cases rose from 33 on May 18, 2026, to 105 on May 24, 2026, and 598 on June 8, 2026.¹⁵ Laboratory-confirmed cases were preferred over suspected cases because they are specific to BDBV and

See Online for appendix 3

Week	Date	Median cumulative confirmed cases	50% PI	90% PI	Deaths
Week 7	May 18, 2026*	294	254–334	212–375	49
Week 8	May 27, 2026	399	340–442	289–502	67
Week 10	June 10, 2026†	648	560–732	470–812	110
Week 12	June 24, 2026	990	858–1141	709–1293	174
Week 16	July 22, 2026	2114	1798–2502	1450–2980	380
Week 20	Aug 19, 2026	4242	3506–5227	2748–6528	742
Week 24	Sept 16, 2026	8210	6636–10 287	5063–13 498	1440

PI=prediction interval. *Week 7 is shown for reference. †Calibration anchor (598 laboratory-confirmed cases on June 8, 2026).

Table 1: Central-transmissibility scenario—recalibrated cumulative reported cases

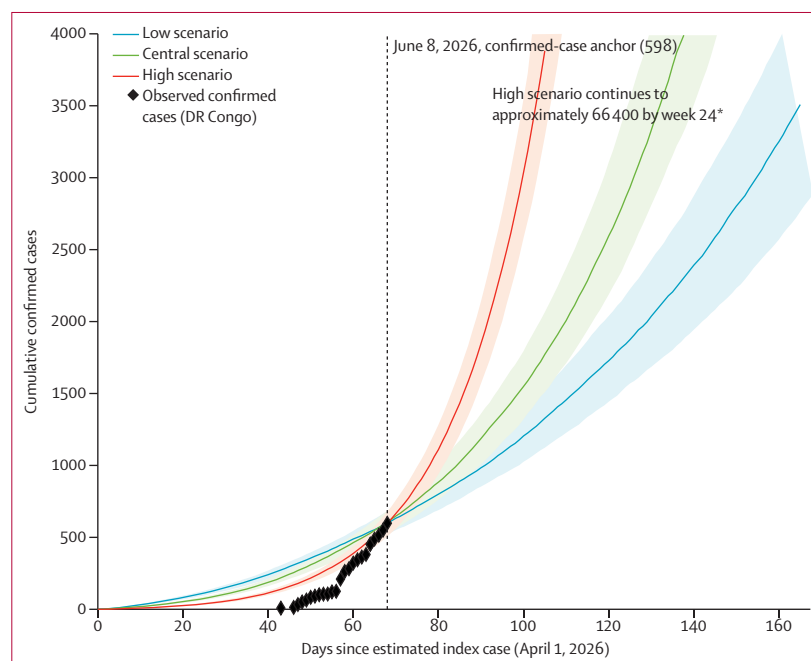


Figure 1: Recalibrated stochastic ensemble projections of cumulative reported BDBV cases (low, central, high scenarios) with observed cumulative trajectory and May 18, 2026, calibration anchor
Scenarios were classified as low (R_0 =1.42); central (R_0 =1.71); and high (R_0 =2.08). BDBV=Bundibugyo virus. R_0 =basic reproduction number. *Off scale.

not subject to the over-ascertainment affecting suspected counts; the early suspected-case series was in any case substantially revised by laboratory reclassification (peaking at 1077 on May 26, 2026, before being resolved to far smaller numbers) and is reported for context only. The model was anchored on the most recent reliable confirmed count (598 on June 8).

Response intensity

The model incorporated a transmission reduction factor representing the combined effects of infection prevention and control measures, community engagement, safe and dignified burial, and contact tracing, with reduction factors varying by scenario to reflect differing operational effectiveness from the date of outbreak declaration onward.

Spillover model

Importation risk to Uganda, South Sudan, Rwanda, and Burundi was modelled as a daily Poisson hazard driven by calibrated infectious prevalence, country-specific cross-border mobility, point of entry detection sensitivity (from WHO Joint External Evaluation assessments), and travel-while-infectious parameters (Uganda 12%; South Sudan 15%; Rwanda and Burundi 8%). Full equations and parameter values are in appendix 3 (pp 2–3). These parameters were not empirically derived for this outbreak and represent the principal sources of uncertainty in spillover projections. Importation probability = $1 - \exp(\text{cumulative 12-week hazard})$. Probabilities of at least one importation were estimated for Uganda, South Sudan, Rwanda, and Burundi.^{16,17}

Sensitivity analyses

Sensitivity analyses examined travel-while-infectious (5–25%), reporting completeness (0.60–0.90), and catchment size (250 000–500 000). Results are in appendix 3 (pp 7–8). These are exploratory analyses characterising directional sensitivity, not formal identifiability assessments.

Role of the funding source

There was no funding source for this study.

Results

Under the central transmissibility scenario (mean accepted $R_0=1.71$), the ensemble calibrated to the confirmed-case anchor projected a median of 648 cumulative confirmed cases by week 10 (June 8, 2026; 50% prediction interval [PI] 560–732), rising to 990 by week 12 (June 24, 2026; 50% PI 858–1141 and 90% PI 709–1293), with a projected median of 174 deaths at week 12 (table 1). Laboratory-confirmed DR Congo cases rose from 33 on May 18, 2026, to 105 by May 24, 2026, and 598 by June 8, 2026; the confirmed-case trajectory is most consistent with the central scenario, with the recent flattening placing the near-term trajectory in the low-to-central range. (For context, the early suspected-case count peaked at 1077 on May 26, 2026, before being substantially revised downward by laboratory reclassification; suspected cases were not used for calibration.) By week 24, the central scenario projected a median of 8210 cumulative confirmed cases (50% PI 6636–10 287 and 90% PI 5063–13 498) and approximately 1440 deaths if transmission were merely sustained. Figure 1 displays the calibrated fan chart with

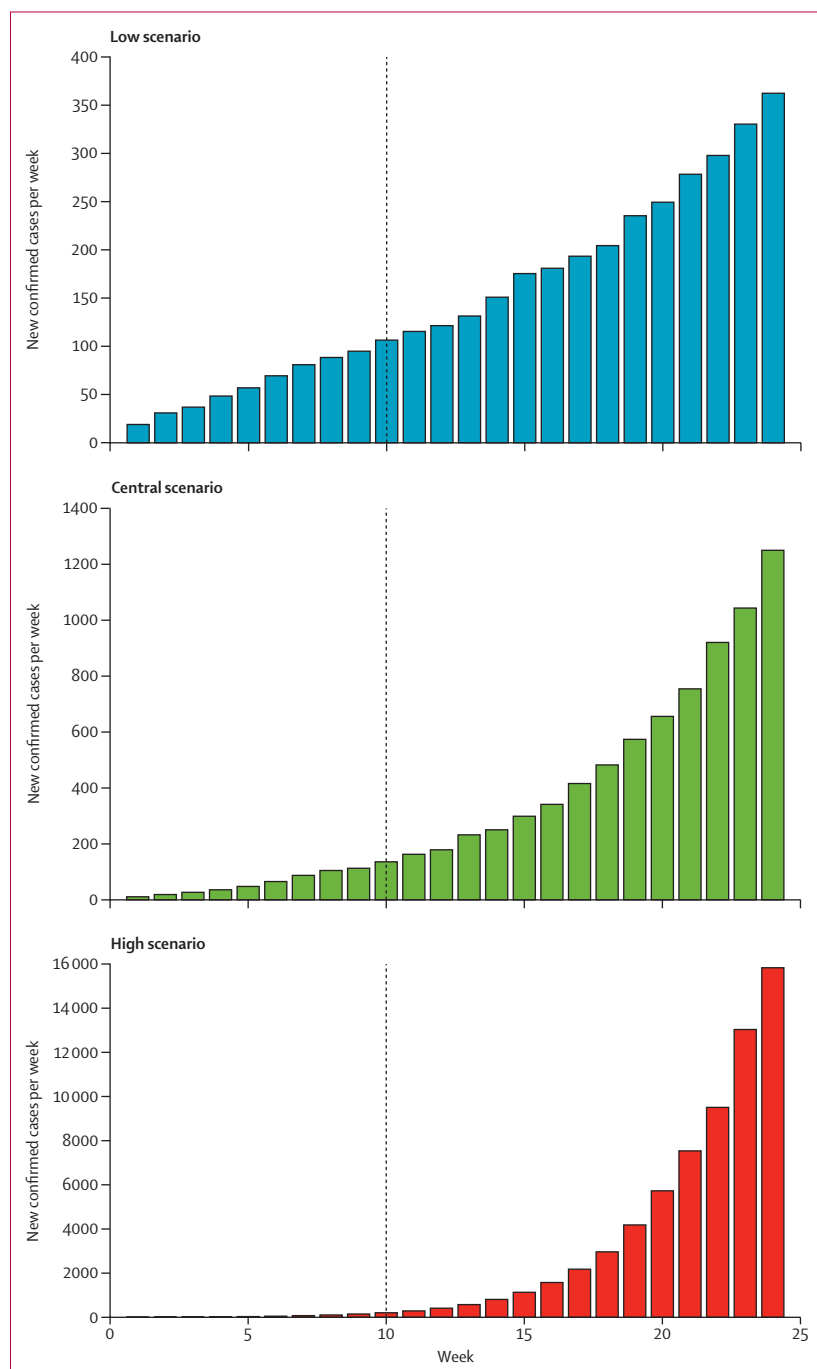


Figure 2: Weekly incidence projections by transmissibility scenario

Incidence projections are anchored at week 10 (June 8, 2026). Note the different scales of infections on the y axes.

the observed confirmed-case trajectory overlaid; figure 2 presents weekly incidence projections by scenario. The central-scenario death trajectory is shown in appendix 3 (p 12).

Under the low-transmissibility scenario (mean accepted $R_0=1.42$), cumulative confirmed cases reached a median of 870 by week 12 (50% PI 752–1004 and 90% PI 641–1133; 160 deaths) and 3662 by week 24 (50% PI 2884–4484 and 90% PI 2094–6037; 673 deaths; appendix p 10). Under the high-transmissibility scenario (mean accepted $R_0=2.08$), confirmed cases reached a median of 1364 (50% PI 1154–1579 and 90% PI 975–1807; 202 deaths) by week 12, rising to 66420 (50% PI 56560–78937 and 90% PI 41718–95881; 10644 deaths) by week 24 (appendix 3 p 9). Because the observed confirmed-case trajectory is consistent with the central scenario, the high scenario is not the operative trajectory; it is retained as an upper-bound contingency should control measures lapse. The accepted R_0 distributions from simulation filtering are shown in appendix 3 (p 11).

Cross-border spillover projections (figure 3 and table 2) indicated very high importation probability for Uganda (94.2%; expected imports 2.85), where the outbreak is now firmly established: as of June 4, 2026, Uganda reported 19 confirmed cases (including five health-care worker infections and two deaths), reflecting sustained importation and onward local transmission since the first imported case. South Sudan had a markedly elevated projected importation probability of 69.3% (expected imports 1.18), approximately double the initial estimate of 37.5%, driven by the larger and more sustained infectious pool spanning Ituri, North Kivu, and South Kivu Provinces, and the persistent porosity of the Aru border. Although the DR Congo confirmed-case trajectory fits with the low-to-central scenario, the cumulative infectious pool already generated keeps near-term spillover risk to South Sudan substantial. Rwanda (8.6%) and Burundi (2.0%) remained at low risk.

Sensitivity analyses confirmed that travel during the infectious period and effective catchment size were the dominant contributors to uncertainty in spillover projections. Varying the travel-while-infectious parameter from 5% to 25% changed Uganda's importation probability by less than 5 percentage points (now saturating at the upper end of the distribution) but altered South Sudan's projected importation probability by approximately 20 percentage points (appendix 3 pp 7–8).

Discussion

The 2026 BDBV outbreak in Ituri Province expanded rapidly during the pre-declaration period and spread across three provinces, and as of June 15, 2026, the outbreak is now closer to the central scenario, with DR Congo confirmed cases rising to 837.¹⁵ The trajectory reflects continued growth consistent with the central scenario rather than a downward bend. Under the central

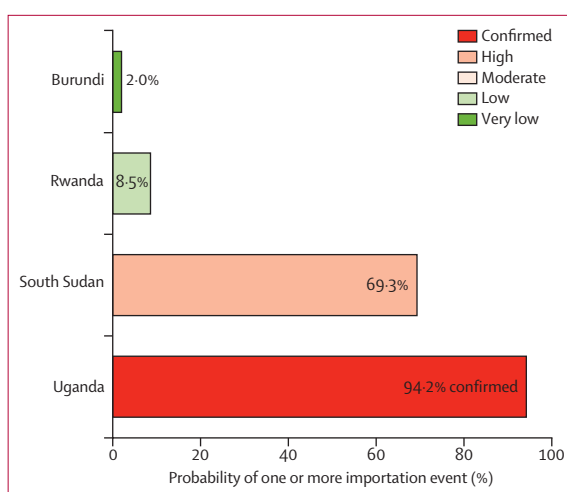


Figure 3: Recalibrated spillover probability to neighbouring countries, 12-week horizon

	Status	Expected imports	Probability of one or more importation event, %	Key drivers
Uganda	Confirmed (×19 as of June 4, 2026; imported and locally acquired)	2.85	94.2%	Mahagi-Goli-Paidha corridor; high mobility. As of June 4, 2026: 19 confirmed cases (imported and locally acquired), including five health-care worker infections and two deaths; sustained importation with onward local transmission; active contact follow-up and district task forces in high-risk districts.
South Sudan	High	1.18	69.3%	Aru border porosity; weakest infection prevention and control and case-management readiness among neighbouring countries; high onward-spread risk if importation occurs. Although the DR Congo trajectory now appears to be bending towards the lower scenarios, the cumulative infectious pool already generated keeps near-term importation risk substantial.
Rwanda	Low	0.09	8.6%	No direct Ituri border; transit via North Kivu; strong national readiness.
Burundi	Very low	0.02	2.0%	No direct border with affected catchment; minimal mobility from Ituri.

Table 2: Recalibrated spillover projections to neighbouring countries (12-week horizon from May 19, 2026)

scenario, the recalibrated ensemble projects a median of 990 cumulative confirmed cases (90% PI 709–1293) and 174 deaths by week 12, calibrated to the laboratory-confirmed anchor of 598 cases on June 8, 2026. The decisive change since the previous submission is the laboratory reclassification of the early suspected case backlog: the suspected count peaked at 1077 on May 26, 2026, and was then sharply revised downward, so the laboratory-confirmed series (598 cases) became the reliable comparator. On this metric the outbreak now tracks at or below the low-to-central projection rather than the high scenario reported previously. The low-scenario trajectory is consistent with successful infection

prevention and control (IPC) intensification; the central scenario requires sustained international support; and the high scenario is retained only as an upper-bound contingency should control lapse. The principal driver of the early escalation was the 6-week pre-declaration interval, which allowed transmission chains to compound silently before any response was mobilised. This underscores the central role of surveillance sensitivity, multisectoral laboratory working groups, and One Health platforms in shortening the pre-detection window during future filovirus outbreaks.

Regional cross-border spillover is an established reality. As of June 14, 2026, in Uganda there were 19 confirmed cases, two confirmed deaths, and one probable death,¹⁵ comprising imported DR Congo nationals and locally acquired Ugandan nationals, including five health-care worker infections and two deaths, which validates the model's 94·2% importation probability. The Mahagi-Goli-Paidha corridor facilitated initial importation; subsequent health-care worker infections in Kampala demonstrate the importance of IPC strengthening in receiving health facilities, a lesson applicable to South Sudan's weaker health system. Uganda's rapid detection and containment response, supported by district task forces activated across high-risk districts and active contact follow-up, reflects the value of previous International Health Regulations (IHR) capacity investments and accumulated Ebola response experience.

This recalibrated analysis sharpens the operational relevance of the IHR framework in shaping national and regional response systems.¹⁸ Uganda's confirmed importation probability (94·2%) and South Sudan's modelled importation probability (69·3%) demonstrate the operational criticality of IHR provisions. The PHEIC declaration and Africa CDC's PHECS provide the institutional architecture to accelerate cross-border preparedness investment under the equity-focused 2024 IHR amendments.^{19–21}

This analysis has implications for response readiness. Although South Sudan's importation probability is lower than Uganda's, documented deficits across all five WHO preparedness domains mean any importation would generate a substantially larger secondary outbreak. Closing these gaps within 4–8 weeks is the principal operational window. Rwanda's effective containment of imported Marburg virus disease, through pre-positioned capacity, rapid contact tracing, and international coordination, offers a directly relevant model for South Sudan's preparedness reinforcement.¹²

Our analysis has some limitations; homogeneous mixing oversimplifies spatial heterogeneity across health zones; health-care worker transmission was not separately modelled; response intensity was applied uniformly; the model is anchored on a single recent confirmed-case datum (598 on June 8, 2026), so the fit is sensitive to that anchor; simulation-filtering is not formal inference (appendix 3 p 6); and informal cross-border

mobility estimates are imprecise. These projections require iterative recalibration as new data emerge.²² The response-intensity factor was applied uniformly from May 15, but real-world response effectiveness varies by place, time, access, and community acceptance. Live mobility data and behavioural adaptation were not incorporated. Surveillance dynamics, particularly the post-declaration case-finding surge, probably contributed to the rapid apparent growth between May 15 and 19; we addressed this by lowering the reporting fraction and widening the calibration window, but residual confounding between transmission and surveillance intensity remains. Most importantly, the calibration relied on early-phase surveillance counts captured in the first weekly WHO External Situation Report (data as of May 18); the cumulative suspected-case figure of 528 was only partially resolved into 35 laboratory-confirmed cases at the time of analysis, and the associated death count of 132 reflected surveillance signals across the affected health zones rather than laboratory-confirmed mortality alone (of which there were five). Misclassification, delayed laboratory throughput, and ongoing active case finding could each shift the calibration anchor in either direction. The projections presented here should therefore be interpreted as an early situational assessment rather than a definitive forecast and will be updated as verified line-list data become available. Future iterations should integrate line lists, contact-tracing data, and dynamic mobility estimates. Individual-level sex and gender data were not available in the aggregate surveillance reports used for this modelling study; disaggregated analysis by sex or gender was therefore not possible. Race and ethnicity data were not collected, as this is a population-level modelling study using aggregate surveillance data.

The 2026 BDBV outbreak in Ituri Province initially escalated faster than anticipated from early surveillance, but the most recent laboratory-confirmed data indicate the trajectory is now bending towards the lower modelled scenarios. After the early suspected case backlog was resolved by laboratory reclassification, the laboratory-confirmed series rose from 105 on May 24, 2026, to 598 by June 8, 2026. The outbreak now tracks at or below the central scenario (mean accepted $R_0=1.71$) rather than the high scenario. Recalibrated central-scenario projections nonetheless indicate continued growth to a median of approximately 990 confirmed cases and 174 deaths by week 12 if current control is merely sustained, underscoring that the gains remain fragile. Spillover is no longer hypothetical: as of June 22, 2026, DR Congo has 1048 confirmed cases and 267 deaths and Uganda has reported 20 confirmed cases, two confirmed deaths, and one probable death,¹⁵ and the estimated probability of importation into South Sudan remains 69·3%.

Contributors

DC conceived the study, led model development, and had full access to all data in the study and final responsibility for the decision to submit

for publication. IFC contributed to spillover model specification, sensitivity analyses, and manuscript revision. All authors contributed to interpretation of findings, and drafting and revision of the manuscript. All authors approved the final version.

Equitable partnership declaration

The authors of this paper have submitted an Equitable Partnership Declaration (appendix 4). This statement allows researchers to describe how their work engages with researchers, communities, and environments in the countries of study. This statement is part of The Lancet Group's broader goal to decolonise global health.

Declaration of interests

We declare no competing interests.

Data sharing

Aggregated modelling outputs, parameter assumptions, and summary tables will be made available upon reasonable request to the corresponding author (Dr Dick Chamla; chamlad@who.int). No individual participant data were collected. Data are derived from publicly available WHO External Situation Reports. Modelling code is implemented within the WHO AFRO PDX platform; access is available to WHO Member States and authorised partners. Aggregated output tables will be available at publication. Code access requests are subject to WHO institutional review.

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